50-Day AI/ML Engineer Interview Prep Checklist

Daily 2 hrs: 45 min coding (25-min time box, then re-code the solution yourself) | 60 min concepts + 5-line notes | 15 min answer the check **out loud, no notes**. Tick only when you can. E/M/H = Easy/Medium/Hard.

Done	Day	Coding problem (E/M/H)	Concepts to learn	Check
	1	Encode & Decode Strings (Hash/string, M)	**ML workflow:** problem framing, train/val/test split, data leakage, baselines	Leakage example
	2	Contains Duplicate (Hash set, E)	**Linear regression:** MSE loss, gradient descent, learning rate, batch vs mini-batch	Explain GD
	3	Valid Anagram (Hash map, E)	**Logistic regression:** sigmoid, log loss, decision boundary, softmax for multi-class	Why log loss?
	4	Group Anagrams (Hash map, M)	**Generalisation:** bias vs variance, under/overfitting, L1 vs L2, early stopping	Fix overfitting
	5	Top K Frequent Elements (Hash map + heap, M)	**Classification metrics:** confusion matrix, precision, recall, F1, ROC-AUC vs PR-AUC, thresholds	Metric for fraud?
	6	Product of Array Except Self (Array, M)	**Validation:** MAE, RMSE, R^2; k-fold, stratified, time-series split; grid vs random search	Pick a split
	7	Longest Consecutive Sequence (Hash set, M)	**Trees:** Gini/entropy, depth; random forest (bagging); gradient boosting, XGBoost	Bagging vs boosting
	8	Subarray Sum Equals K (Prefix sum + hash, M)	**Features:** scaling, one-hot vs target encoding, missing values, class weights, SMOTE	3 imbalance fixes
	9	Valid Sudoku (Hash set, M)	**Unsupervised:** k-means, elbow method, PCA, explained variance	PCA simply
	10	Valid Palindrome (Two pointers, E)	**Neural nets:** activations (ReLU, sigmoid, softmax), forward pass, backprop, vanishing gradients	Why ReLU?
	11	3Sum (Two pointers, M)	**Training:** SGD, momentum, Adam, LR schedules, weight init, batch norm, dropout	Role of dropout
	12	Container With Most Water (Two pointers, M)	**SQL 1:** SELECT/WHERE, INNER/LEFT/SELF joins, GROUP BY + HAVING, NULL handling	Join types
	13	Best Time to Buy/Sell Stock (Sliding window, E)	**SQL 2:** subqueries, CTEs, window functions (ROW_NUMBER, RANK, LAG, running SUM), indexes	Top 3 per group
	14	Longest Substring w/o Repeating (Sliding window, M)	**Sequence models:** CNN basics, RNN/LSTM, why transformers replaced RNNs	RNN limits
	15	Longest Repeating Char Replacement (Window, M)	**Text to vectors:** tokenisation (BPE, WordPiece), embeddings, cosine similarity	Why close vectors?
	16	Valid Parentheses (Stack, E)	**Attention:** Q, K, V, scaled dot-product, multi-head, positional encoding, causal mask	Attention in 2 min
	17	Min Stack (Stack, M)	**Transformer block:** residuals, layer norm, FFN; BERT vs GPT vs T5	Which for what?
	18	Daily Temperatures (Monotonic stack, M)	**LLM training:** pretraining, SFT, RLHF vs DPO, context window, scaling	SFT vs RLHF
	19	Binary Search (E)	**Decoding:** greedy, beam search, temperature, top-k, top-p, KV cache	Temperature
	20	Search in Rotated Sorted Array (Binary search, M)	**Adapting LLMs:** prompting vs RAG vs fine-tuning, LoRA, QLoRA, PEFT	FAQ bot choice
	21	Find Min in Rotated Array (Binary search, M)	**Prompting:** system prompts, few-shot, chain-of-thought, JSON output, prompt injection	Fix a vague prompt
	22	Reverse Linked List (E)	**RAG pipeline:** ingest, parse, chunk, embed, index, retrieve, augment, generate	Draw RAG
	23	Merge Two Sorted Lists (Linked list, E)	**Chunking:** fixed, recursive, semantic, structure-aware; size, overlap, metadata	Chunk size trade-off
	24	Linked List Cycle (Fast/slow, E)	**Vector DBs:** FAISS, Chroma, pgvector, Pinecone; cosine vs dot vs L2; ANN, HNSW, IVF	Why ANN?
	25	Remove Nth Node From End (Linked list, M)	**Better retrieval:** BM25, hybrid search, RRF, cross-encoder reranking, query rewriting, HyDE	Fix bad retrieval
	26	LRU Cache (Hash map + linked list, M)	**LLM evals 1:** golden datasets, exact match, BLEU/ROUGE limits, LLM-as-judge, human review	Build an eval set
	27	Invert Binary Tree (E)	**LLM evals 2 (RAG):** recall@k, MRR, faithfulness, answer relevance, context precision, RAGAS, regression tests	Is RAG good?
	28	Max Depth of Binary Tree (DFS, E)	**Hallucination:** causes, grounding, citations, abstaining, guardrails, output validation	3 fixes
	29	Same Tree (E)	**Agents:** tool calling, ReAct loop, planning, stop conditions, tool error handling	How tools are picked
	30	Level Order Traversal (BFS, M)	**Agent design:** short/long-term memory, multi-agent, LangGraph state, human-in-the-loop, MCP	When multi-agent?
	31	Validate BST (M)	**Latency:** TTFT vs throughput, streaming, batching, KV cache, quantisation (INT8/4), semantic cache	Cut latency
	32	LCA of BST (M)	**Cost + observability:** tokens, prompt caching, model routing, context trimming, tracing (LangSmith)	Cut cost
	33	Kth Largest Element (Heap, M)	**APIs:** REST verbs, status codes, FastAPI, Pydantic validation, async/await, SSE streaming	Request lifecycle
	34	Merge K Sorted Lists (Heap, H)	**Docker 1:** image vs container, Dockerfile, layers & caching, build/run, ports, volumes, env vars	Image vs container
	35	Number of Islands (Graph DFS/BFS, M)	**Docker 2:** multi-stage builds, slim images, docker-compose, registries, secrets	Why compose?
	36	Clone Graph (Graph + hash map, M)	**CI/CD:** Git branches & PRs, pytest, GitHub Actions: lint → test → build → deploy, staging vs prod	Draw a pipeline
	37	Course Schedule (Topological sort, M)	**MLOps:** MLflow tracking, model registry, data/model versioning, monitoring, data vs concept drift, retraining	Drift types
	38	Rotting Oranges (Multi-source BFS, M)	**Design framework:** functional & non-functional reqs, estimates, API, high-level, deep dive, trade-offs	Recite steps
	39	Subsets (Backtracking, M)	**Scaling:** vertical vs horizontal, load balancing, stateless services, Redis cache (TTL, LRU), CDN	Cache placement
	40	Combination Sum (Backtracking, M)	**Data layer:** SQL vs NoSQL, indexing, replication, sharding, consistency, CAP theorem	CAP example
	41	Word Search (Backtracking, M)	**Async systems:** Kafka/SQS, pub/sub, workers, retries, idempotency, rate limiting (token bucket)	Why a queue?

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	42	Climbing Stairs (DP, E)	**DESIGN:** URL shortener (ID generation, redirects, caching, analytics)	30 min on paper
	43	House Robber (DP, M)	**DESIGN:** chat / notification system (WebSockets, fan-out, delivery guarantees)	1M users online
	44	Coin Change (DP, M)	**ML DESIGN:** recommender (candidate gen, ranking, features, offline/online metrics, A/B tests)	A/B test plan
	45	Longest Increasing Subsequence (DP, M)	**ML DESIGN:** model serving (batch vs real-time, autoscaling, GPUs, canary rollout, monitoring)	Batch vs real-time
	46	Word Break (DP + hash set, M)	**AI DESIGN:** enterprise RAG chatbot (access control, doc freshness, evals, cost, latency)	Hourly doc updates
	47	Unique Paths (2D DP, M)	**AI DESIGN:** support agent (tools, guardrails, escalation to human, logging, evals)	Stop unsafe actions
	48	Maximum Subarray (Kadane, M)	**Debugging:** reproduce, isolate, hypothesise, test, fix, verify; ML (loss stuck) & RAG (bad context)	RAG wrong: check first
	49	Merge Intervals (Intervals, M)	**Coding agents:** give context + constraints, plan first, review diffs, write tests, verify output	Your workflow
	50	Implement Trie (Trie, M)	**Behavioural:** 6 STAR stories (challenge, failure, conflict, leadership, fast learning, impact)	Each under 2 min
BONUS (Days 51-60): after Day 50 or in your optional third hour				
	51	Search a 2D Matrix (Binary search, M)	**Probability & stats:** distributions, Bayes, CLT, p-values, Type I/II errors	Bayes example
	52	Koko Eating Bananas (Binary search, M)	**A/B testing:** sample size, power, peeking, guardrail metrics, novelty effect	Design an A/B test
	53	Permutations (Backtracking, M)	**Linear algebra & calculus:** dot product, matrix shapes, eigenvectors/SVD, chain rule	Attention in maths
	54	Generate Parentheses (Backtracking, M)	**Python deep dive:** GIL, asyncio vs threads vs processes, decorators, dataclasses	When asyncio?
	55	Pacific Atlantic Water Flow (Graph, M)	**Networking:** DNS, TCP vs UDP, HTTP/HTTPS, TLS, JWT, CORS	Type a URL: what happens
	56	Minimum Window Substring (Window, H)	**Cloud (AWS):** EC2/ECS/Lambda, S3, IAM, VPC, Bedrock, SageMaker	Deploy a RAG API
	57	Insert Interval (Intervals, M)	**Kubernetes:** pods, deployments, services, ingress, HPA, probes	How K8s scales
	58	Non-overlapping Intervals (Greedy, M)	**Data pipelines:** ETL vs ELT, Airflow DAGs, lake vs warehouse, quality checks	Nightly pipeline
	59	Decode Ways (DP, M)	**LLM security:** OWASP LLM Top 10, indirect injection, PII, excessive agency	3 defences
	60	Longest Common Subsequence (2D DP, M)	**Multimodal:** ViT, CLIP, vision-language models, multimodal RAG, voice agents	Explain a VLM

Order: 1-11 ML, 12-13 SQL, 14-21 Transformers & LLMs, 22-32 RAG, evals, agents, latency, 33-37 engineering, 38-47 system design, 48-50 debugging & behavioural, 51-60 bonus. **Resources:** NeetCode 150, StatQuest, Illustrated Transformer, Chip Huyen's book, ByteByteGo.